

Applied Class 3 – Exercises

A paper appearing in a 2015 issue of *Biological Conservation* used a randomised response technique to investigate rates of illegal fishing of red abalone in Northern California. The study involved interviews with people at various sites. Once verbal consent to participant was obtained, each respondent was given a coin and an envelope containing two cards. On one card was written the question

In the past year have you ever taken abalone under the minimum legal-size limit?

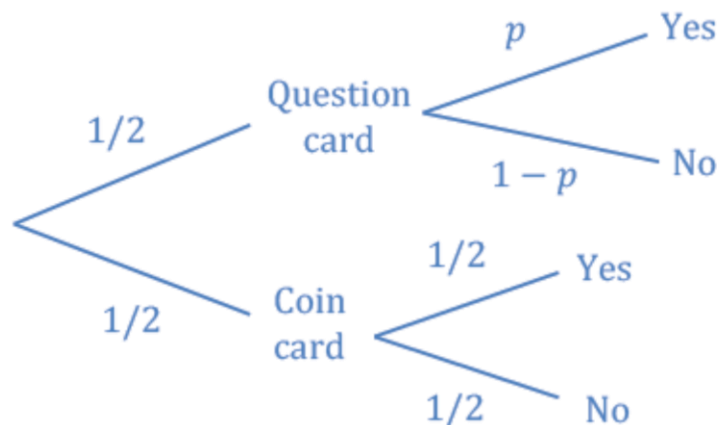
while on the other card was written the question

Did you get heads on the coin toss?

Without the interviewer watching, each respondent first tossed the coin, noting the outcome. They then chose a card at random and answered the question on it to the interviewer. Out of 279 respondents, 102 answered 'Yes'.

- a) Denoting the unknown proportion by p , draw a tree diagram showing the conditional probabilities of answering 'Yes' following this procedure.

Note that it is easier to branch on the choice of card, even though the coin toss happens first:



The total probability of 'Yes' is $P(Yes) = \frac{1}{2} \times p + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}p + \frac{1}{4}$.

- b) Based on your tree diagram in (a), what is the estimated proportion of people who have actually taken abalone under the minimum legal-size limit in the last year?

We estimate $P(Yes) \approx \frac{102}{279} = 0.366$, so $\frac{1}{2}p + \frac{1}{4} \approx 0.366$.

Rearranging gives $\frac{1}{2}p \approx 0.366 - \frac{1}{4} = 0.116$, so $p \approx 2 \times 0.116 = 0.232$.

Thus, we estimate 23.2% of people have taken abalone under the minimum legal-size limit.